

Problem Chosen: Smart Health Solutions for Solapur Municipal Corporation.

Problem Definition

Problem context

Solapur Municipal Corporation serves a densely populated urban population with significant public-health vulnerability, including communicable disease risk, maternal-child health challenges, and infrastructure stress. Frontline surveillance in the city relies heavily on ASHA workers and healthcare facilities, yet health data is fragmented across paper registers, isolated applications, and disconnected hospital reporting systems.

The absence of a centralized electronic health data platform and real-time analytics prevents proactive public health planning. Limited interoperability, lack of standardized data formats, and non-FHIR-aligned reporting further restrict effective syndromic surveillance. This gap significantly impacts emergency preparedness, resource optimization, and municipal health governance.

Objectives

1. Enable **real-time ingestion** of standardized field reports (ASHA) and facility data (hospitals, labs) at ward granularity.
2. Provide **ward-level surveillance** and anomaly detection so SMC can detect rising symptom clusters or resource stress early.
3. Deliver a municipal **command dashboard** for evidence-driven decisioning (heatmaps, capacity views, alert queues).
4. Offer a **citizen interface** for localized advisories and facility availability (targeted push notifications).
5. Preserve **data security, privacy, and local operability** — runs on municipal infrastructure or government cloud and supports offline field capture.

Constraints

The solution must address strict data security, privacy compliance under the DPDP Act, and encrypted communication using TLS and JWT-based authentication

Integration challenges exist due to heterogeneous hospital systems and limited standardized reporting formats. Connectivity limitations in field environments require offline-first architecture with synchronization mechanisms. Infrastructure scalability, resource constraints, and interoperability with national systems such as ABDM and IDSP must also be considered.

Functions

The system performs health data aggregation, validation, and FHIR-based standardization through a secure API ingestion layer.

It delivers real-time analytics dashboards with ward-level heatmaps, temporal trend analysis, and facility resource stress monitoring. The Analytics & Alert Engine conducts predictive disease surveillance using anomaly detection and threshold-based models. Additionally, it provides zone-wise risk identification, automated alert dissemination via FCM, and public health advisory access through citizen-facing interfaces.

Early-Stage Solution Architecture

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List of Subsystems – Arogya-SMC

Sl.No	Subsystem Name	Functionality
1.	ASHA Field Reporting Interface	Tech Stack: <i>Flutter · SQLite · FCM</i> Capabilities: • Offline syndromic indicator capture (fever, cough, diarrhoea) • Maternal and child risk flag reporting • Geotagged environmental health observations • Background synchronization with central server • Low-bandwidth resilient submission workflow Data Exchange: HTTPS (JSON, JWT) Standards: FHIR Observation (SNOMED CT / ICD-11)
2.	Citizen Health Access & Engagement App	Tech Stack: <i>Flutter · GPS · FCM</i> Capabilities: • Ward-aware facility and bed availability lookup • Public health advisories and outbreak alerts • Consent-based location access with offline cache • Citizen notification delivery via push • Nearest facility routing and contact access Data Exchange: Public HTTPS (JSON) Security: No authentication (public access)
3.	Hospital Capacity Reporting Portal	Tech Stack: <i>React · TypeScript · RBAC</i> Capabilities: • Bed, ICU, ventilator and oxygen capacity reporting • Disease admission and occupancy updates • Historical submission and audit logging • Secure role-based hospital access • Structured daily health reporting interface Data Exchange: HTTPS POST (JSON, JWT) Security: RBAC + optional 2FA

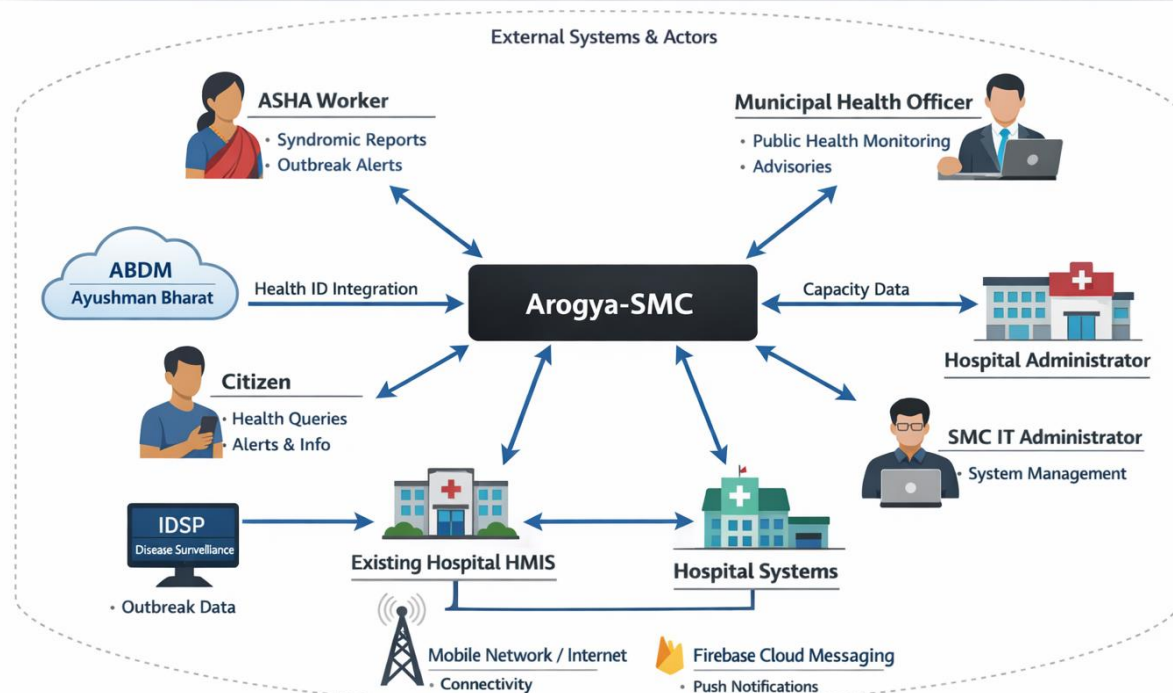
4.	Municipal Health Command Dashboard	Tech Stack: <i>React · Leaflet · Recharts</i> Capabilities: • Ward-level heatmaps and trend analytics • Facility resource stress visualization • Alert monitoring and outbreak oversight • Drill-down epidemiological analysis • Temporal trend and comparison dashboards Data Exchange: HTTPS (JWT) Data Source: Aggregated analytics APIs
5.	Secure API Gateway & Ingestion Layer	Tech Stack: <i>Node.js · Fastify</i> Capabilities: • Unified entry point for all external requests • JWT validation and rate limiting • JSON schema validation and routing • Citizen public endpoint exposure • Service orchestration and request logging Security: TLS 1.3 + JWT Routing: FHIR service / database
6.	FHIR-Compliant Data Standardization Service	Tech Stack: <i>Node.js · FHIR mapping</i> Capabilities: • Raw report transformation to FHIR Observations • SNOMED CT and ICD-11 code alignment • Ward and facility metadata enrichment • Interoperability validation layer • Health data normalization and structuring Output: JSONB FHIR resources Compliance: ABDM / IDSP aligned
7.	Centralized Health Data Repository (PostgreSQL/PostGIS)	Tech Stack: <i>PostgreSQL · PostGIS · JSONB</i> Capabilities: • Standardized observations and capacity storage • Ward spatial boundaries and facility registry • Geospatial queries for heatmaps • Time-series indexed health analytics • High-volume structured health data retention Data Model: FHIR JSONB + relational tables Queries: SQL / spatial

8.	Predictive Analytics & Alert Engine	Tech Stack: <i>Python · Pandas · Prophet · Airflow</i> Capabilities: • Ward-level aggregation and anomaly detection • Temporal outbreak trend modelling • Threshold-based alert generation • Real-time alert publishing via Redis • Predictive risk scoring and forecasting Models: Isolation Forest / time-series Output: Alerts + risk metrics
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➤ Arogya-SMC Core Subsystems

The platform is decomposed into eight modular subsystems, each designed with clear boundaries and well-defined interfaces. Field-facing modules (ASHA Reporting, Citizen App) leverage Flutter for cross-platform delivery and offline-first operation. Administrative interfaces (Hospital Portal, Municipal Dashboard) are built with React and visualisation libraries (Leaflet, Recharts) to provide real-time situational awareness. The backend integrates a secure API Gateway (Node.js/Fastify), a FHIR-compliant standardisation service, and a PostgreSQL/PostGIS data repository that stores standardised observations, capacity metrics, and geospatial ward boundaries. A Python-based analytics engine performs predictive surveillance and automated alerting. All external communications use TLS 1.3, with JWT-based authentication for protected endpoints and FCM for push notifications. Together, these subsystems form a cohesive, standards-aligned ecosystem that addresses Solapur's fragmented health data, delayed outbreak detection, and limited citizen engagement.

Figure 2: Arogya-SMC System Context Diagram – High-level view of the system boundaries, external actors, and neighbouring systems.





Subsystems Interaction matrix

From \ To	ASHA Field Surveillance Module	Citizen Health Access & Engagement App	Hospital Capacity & Disease Reporting Portal	Municipal Health Command Centre	Secure API Gateway & Ingestion Layer	FHIR-Compliant Data Standardization Service	Centralized Health Data Repository (PostgreSQL /PostGIS)	Predictive Analytics & Alert Engine
ASHA Field Surveillance Module	–	–	–	–	Syndromic reports (HTTPS, JSON, JWT); advisory requests (GET)	Raw validated health data	–	–
Citizen Health Access & Engagement App	–	–	–	–	Public queries for facilities/advisories (HTTPS, JSON, no auth)	–	–	–
Hospital Capacity & Disease Reporting Portal	–	–	–	–	Capacity reports (HTTPS, JSON, JWT); history requests (GET)	–	–	–
Municipal Health Command Centre	–	–	–	–	Dashboard queries for trends/heatmaps/alerts (HTTPS, JWT)	–	–	
Secure API Gateway & Ingestion Layer	Push notifications (FCM); Advisory responses (HTTPS)	Push notifications (FCM); public data responses (HTTPS, JSON)	Submission confirmations; history data (HTTPS, JSON)	Dashboard data (HTTPS, JSON)	–	Raw reports for standardization (HTTP / message queue)	Read queries for public & dashboard data (SQL)	



FHIR-Compliant Data Standardization Service	–	–	–	–	–	–	Writes standardized FHIR Observations (JSONB); reads master data (wards, facilities) via SQL	
Centralized Health Data Repository (PostgreSQL /PostGIS)	–	–	–	–	Query results (JSON) via internal API	Master data responses (wards, facilities) via SQL	–	Bulk data exports for analytics (SQL)
Predictive Analytics & Alert Engine	–	–	–	–	Publishes real-time alerts to message queue (Redis) for push delivery	–	Writes alerts & aggregated metrics (SQL); Reads historical data for model training (SQL)	–

➤ Description

The Subsystem Interaction Matrix defines all directed data exchanges across the Arogya-SMC architecture, linking field interfaces, integration layers, data platforms, analytics services, and citizen channels. Each cell identifies the initiating subsystem (row) and receiving subsystem (column), together with the governing protocol (HTTPS, SQL, FCM), data format (JSON, FHIR-aligned, JSONB), authentication context (JWT/public), and interaction purpose.

By explicitly mapping ingestion, standardization, storage, analytics, and dissemination flows, the matrix establishes a clear contract for interoperability, security, and scalability across municipal health data operations. It serves as the authoritative reference for system integration and implementation of Arogya-SMC.



Architecture Diagram

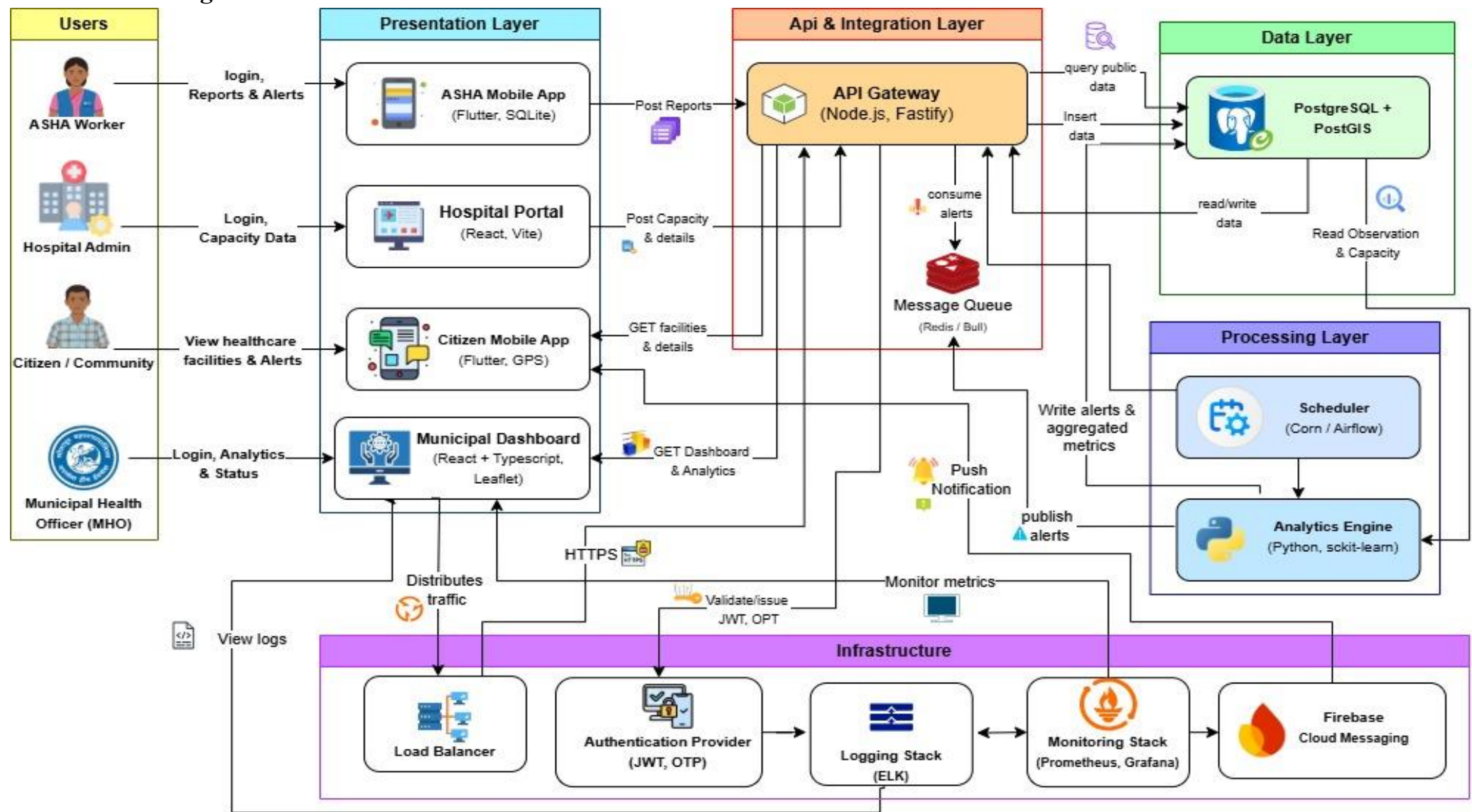


Figure 1: Complete Arogya-SMC System Architecture – Integrated View of All Subsystems and Data Flows

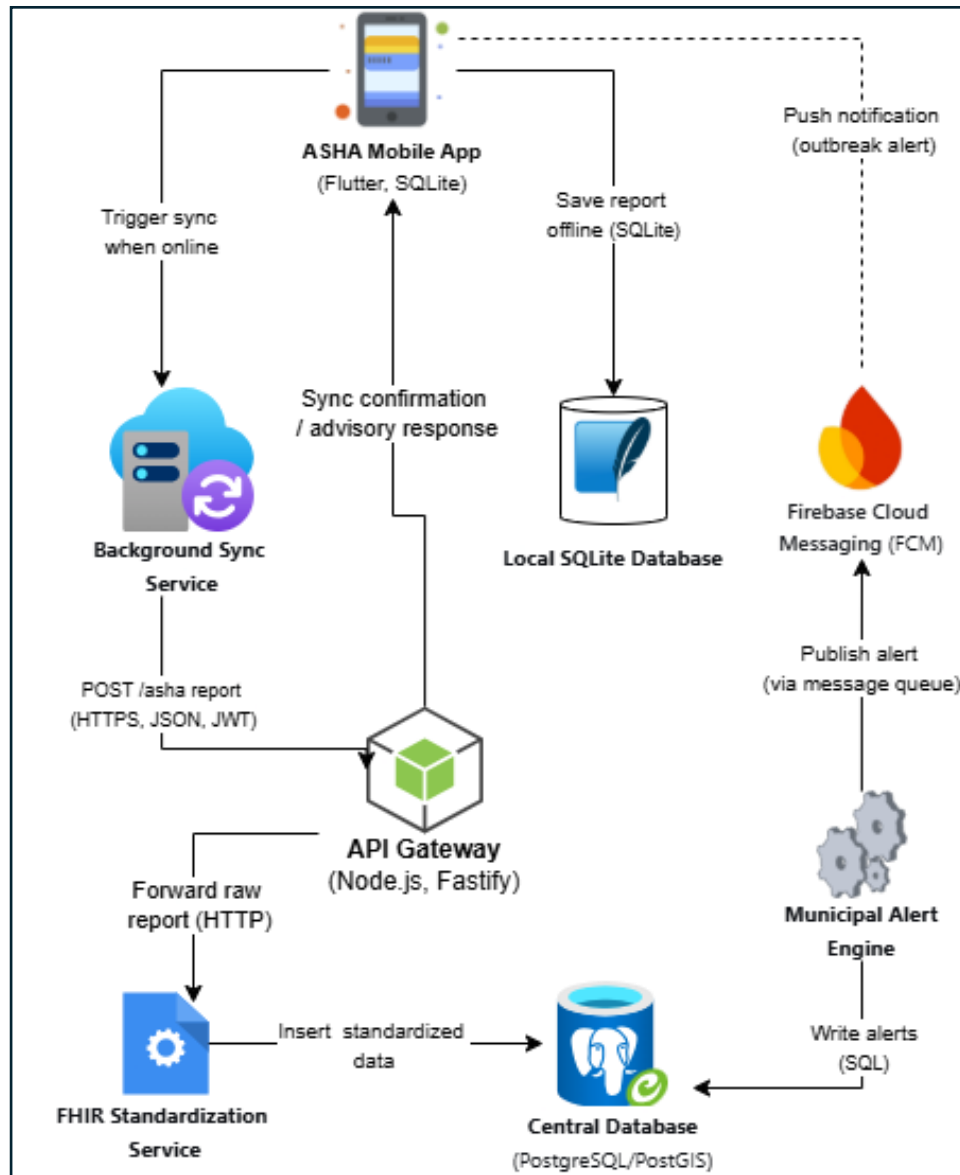


Figure 2: ASHA Field Worker Subsystem –
Offline Data Capture, Synchronization, and Alert Delivery

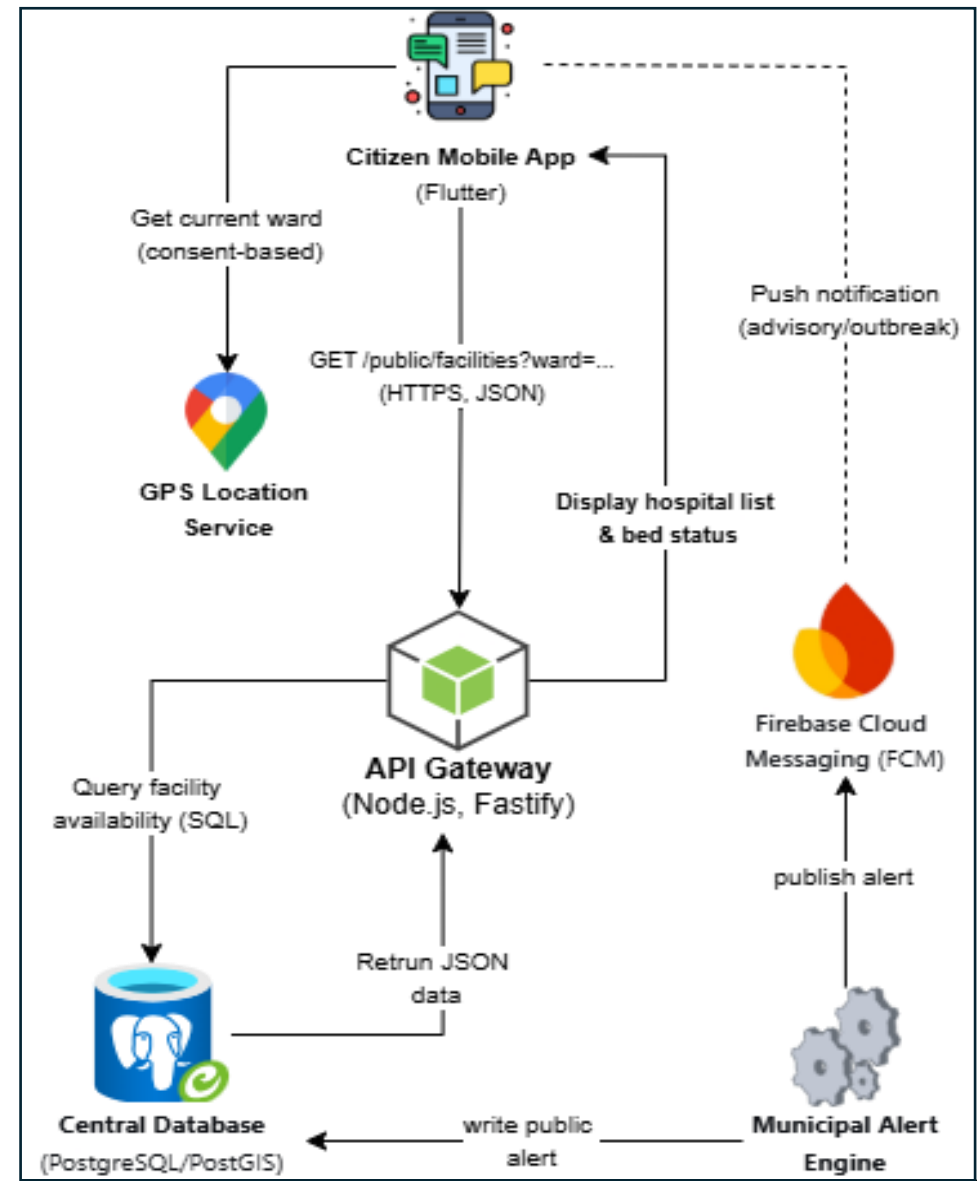


Figure 3: Citizen Engagement Architecture –
Public Access to Real-Time Health Information

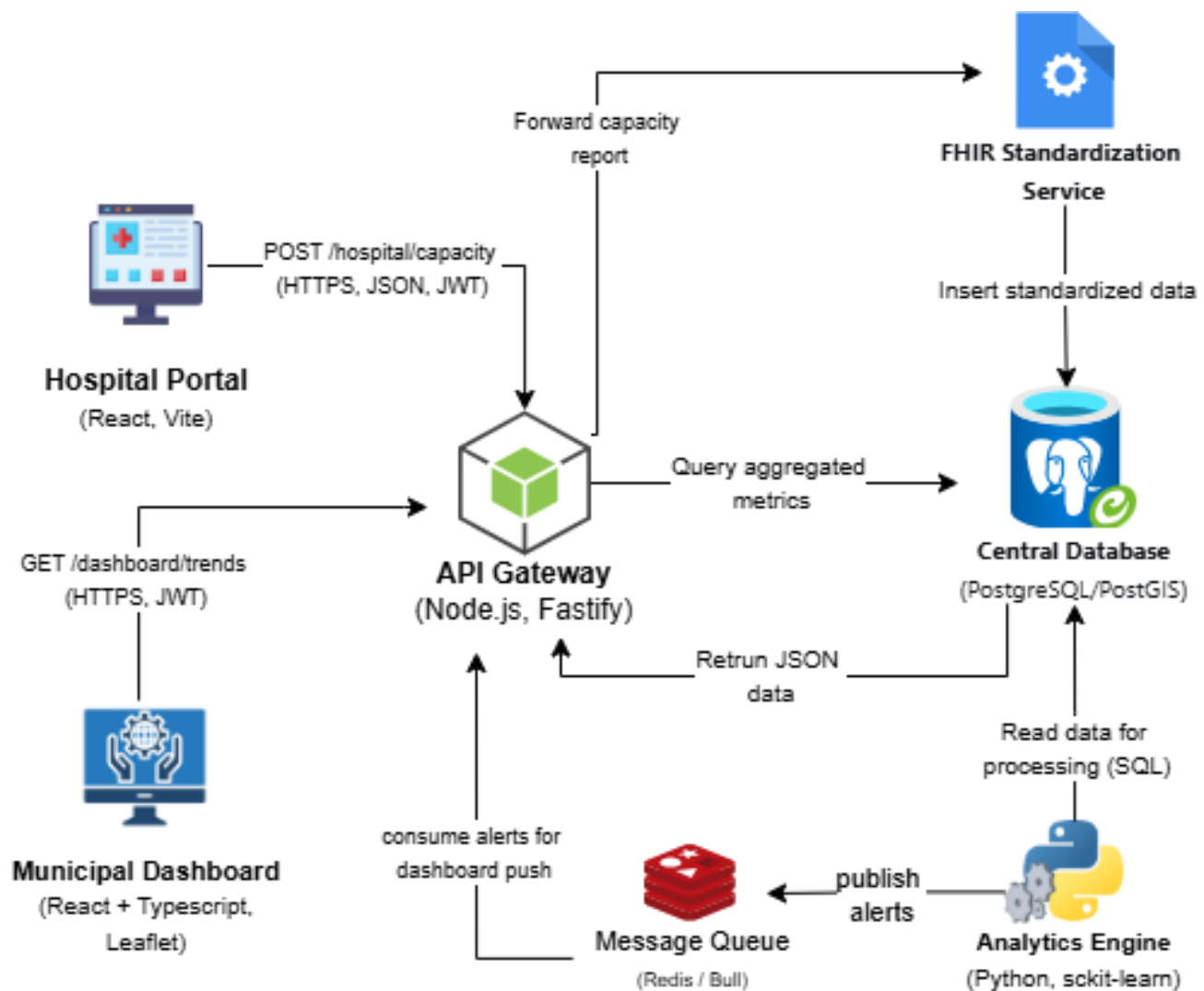


Figure 4: Hospital & Municipal Administration Subsystem – *Capacity Reporting and Decision Support*

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Note: The above details will be used for certificate generation. Please ensure accuracy.

Declaration

We hereby declare that the work titled “Smart Health Solutions for Solapur Municipal Corporation.” submitted for SAMVED-2026 is the original outcome of our team’s efforts.

The work has been carried out by us under the guidance of the undersigned mentor and is **neither copied from any source nor generated using artificial intelligence (AI) tools.**

Place: Vashi

Date: 17/02/2026

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